

systems and computer-brain interface (CBI), a precise brain stimulation technique. These technologies are combined to perform computer-mediated brain-to-brain (B2B) communication.

The information gets transmitted between two minds in the form of binary streams of encoded words. Researchers say that this new 'hyper-interaction' technology will impact the cognitive research in the field of neuroscience and scientific study of consciousness.

Soft robots that can change shape

A new kind of untethered, soft robot is being developed by a team of researchers from Harvard and Cornell University. This robot is able to walk using four legs, and when it commutes its shape also changes. This autonomous robot has some awesome features including its ability to operate through puddles of water or a snowstorm and can even bear limited exposure to heat. If it is run over by an automobile, even then the robot 'feels' fine.



The soft robot

A paper titled 'A Resilient, Untethered Soft Robot' has been published by Wyss Institute for Biologically Inspired Engineering, Department of Chemistry and Chemical Biology, at Harvard University and the School of Mechanical and Aerospace Engineering at Cornell University, which details their research work. Innovative composite materials are being used by researchers in fabricating this soft robot, like silicone elastomer, polyaramid fabric and hollow glass micro-spheres.

The robot is able to carry miniature air compressors, battery pack, valves and controller, and is able to facilitate its autonomous operation. Some new fabrication techniques have been de-

veloped, too, to create its soft body with modified Pneu-Net actuators. The robot's body is so designed that it stays resilient to adverse environmental conditions, as mentioned earlier.

Laser optical tweezers to study how Malaria infects

Malaria, the disease that spreads through mosquitoes, affects millions of people every year and also causes many deaths all over the world. Malaria is caused by *Plasmodium Falciparum* parasite which invades red blood cells one by one. The parasite blocks one by one all the capillaries that transmit blood to the brain and other organs. This invasion and infection process is very swift.

Researchers hardly get to study the infection stage. It is important to study the infection and invasion process in development of effective drugs and vaccines. The researchers from Cambridge University have developed laser optical tweezers to study this interaction process between parasite and red blood cells. They have been studying invasion process but it is important to study pathogenesis process too. Pathogenesis is the development of Malaria. The parasite that causes disease infects red blood cells but it invades in less than a minute after its release.

Laser optical tweezers can record and control the movement of cells. Extremely focused laser beam is used to apply very small force to control cells more precisely. The optical tweezers can be used to select parasites that have just invaded from red blood cells. This method of invasion process is useful for scrutiny as it helps scientists to study the parasite before it loses its potential to infect other cells. Researchers also focused on measuring the adherence between parasites and red blood cells, which turned out to be very weak. This adherence can be blocked using drugs and antibodies.

Laser pulse that can turn glass into metal

Vienna University of Technology's group of researchers have made the

impossible possible by turning glass into metal through ultra-short laser. Hence, a huge change is happening in the science of material properties and that too at a very fast pace.

Scientists from the Vienna University, Georg Wachter, Christoph Lemell and Professor Joachim Burgdorfer, collaborated with the research team of Tsukuba University in Japan to conduct the experiment jointly. In the experiment, electric current was measured in quartz glass while being illuminated by a laser pulse. Once the laser was turned off the material returned to its original state. In this case the laser pulse worked as a strong electrical field. The field was so powerful that it could completely change the structure of electron states of a material.

The project involved several quantum processes which keep happening together. Supercomputers have been of great help in this project to make the interaction between the laser and electron and also among all the electrons. Within modern-day transistors a large number of charge carriers move during each switching operation. Now, with this new experiment, we have come to know that laser pulses are able to switch electric currents thousand times faster than the transistors.

16-year-old invents device to turn breath into speech

A 16-year-old boy from Panipat (Haryana) has created an innovative contraption which is capable of turning breath into speech using Morse code. The young talent, named Arsh Shah Dilbagi, is the only finalist from Asia who has been selected for Google Science Fair, 2014. Arsh has named his innovation, Talk.

Arsh says the device is aimed at helping speech-impaired and paralysed people in their communication process. According to him, this is the fastest and cheapest gadget of the kind.

The device uses signals from a person's breath, using Morse code. The signals are picked up by a sensor and converted into speech. Talk, the device, is categorised under augmentative and alternative communication (AAC) devices and costs about 5000 rupees.

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